

ELEMENTARY CONSERVATION OF WATER

MODULE - 5

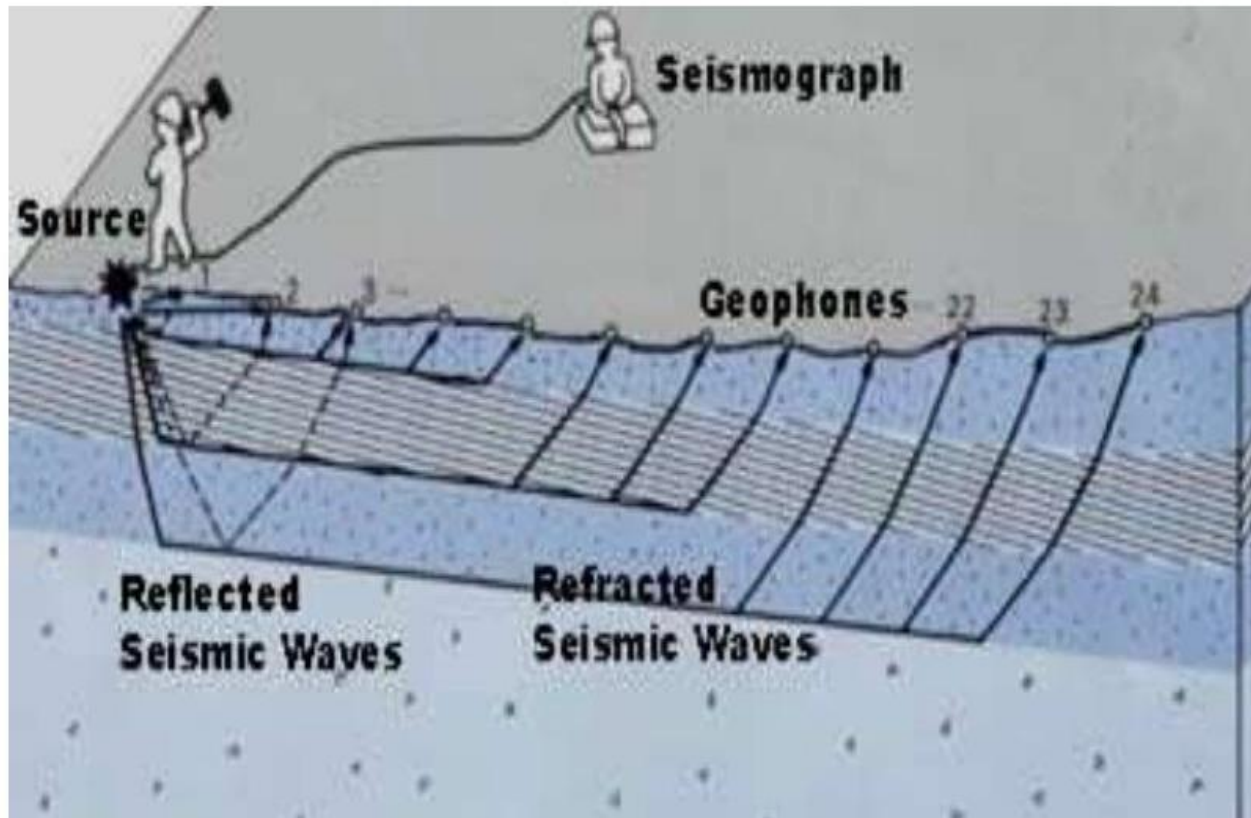
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Sub Surface investigation of Ground Water

- Geophysical methods:
 - Seismic refraction method
 - Electrical resistivity method

Seismic Refraction Method



Principle:

This method is based on the measurement of **travel times of seismic waves** refracted at the interfaces between subsurface layers of different densities.

- A seismic source (hammer blow, explosive, etc.) generates waves.
- Waves travel through the ground and are refracted at layer boundaries.
- Geophones detect the refracted waves at the surface.
- Travel time vs. distance is plotted to identify layer depths and velocities.

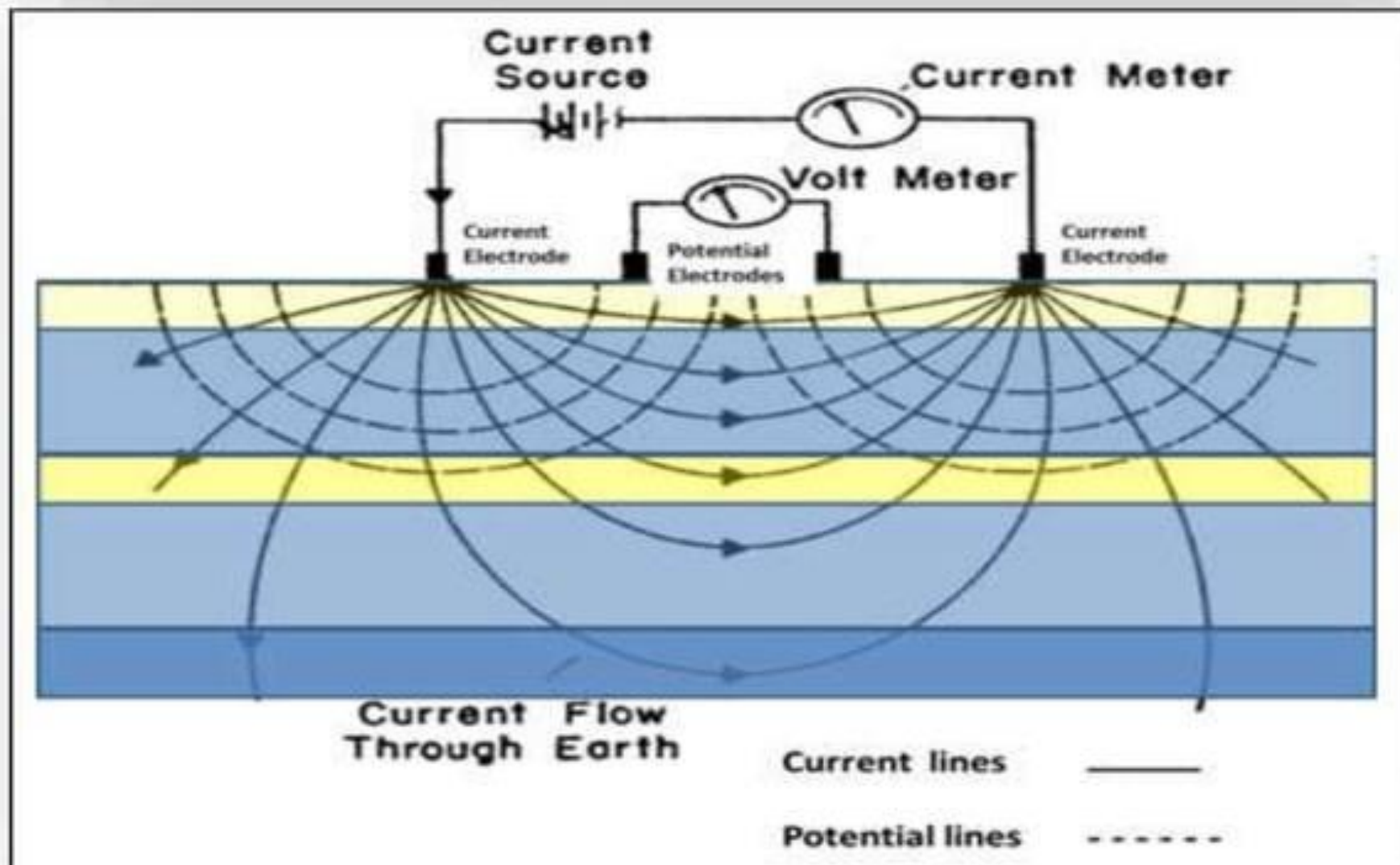
Applications:

- Determining depth to bedrock
- Mapping subsurface layers
- Engineering site investigations
- Groundwater and landslide studies

Advantages:

- Simple and cost-effective
- Provides seismic velocity, which correlates with material type and strength

Electrical Resistivity method



Principle:

This method measures the **resistance of the ground** to the flow of electrical current. Different subsurface materials (e.g., clay, sand, rock, water) have different resistivity values.

- Electrodes are placed in the ground (usually in a line).
- A current is injected into the ground through a pair of electrodes.
- Voltage is measured between another pair of electrodes.
- Apparent resistivity is calculated and used to interpret subsurface structure.

Applications:

- Groundwater exploration
- Detecting contamination plumes
- Mapping bedrock, faults, and cavities
- Geotechnical and archaeological surveys

Advantages:

- Non-destructive and environmentally friendly
- Good resolution for identifying water-bearing zones or voids